

Grokking Research Program

A three-paper collaboration opportunity in mathematical learning, scaling and mechanistic interpretability

PAPER 1 DLP Grokking at Scale	PAPER 2 Representation Discovery	PAPER 3 ECDLP Grokking & Scaling
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Why this collaboration could be useful

Core opportunity: use symbolic regression as a mechanistic-interpretability tool. Instead of only probing whether a latent variable is decodable, search for compact symbolic equations that describe what intermediate activations are computing, how those equations change across checkpoints, and whether the same mathematical rule reappears across model sizes and problem representations.

Paper 1 - Scaling Grokking on the Discrete Logarithm Problem

Scientific question. How do model capacity, training data and group size determine whether a transformer moves from memorization to algorithmic generalization on a difficult inverse algebraic problem?

427k anchor transformer	99.8% test accuracy at p=113	90k-14.4M capacity sweep	p=521 larger-prime grokking
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- **Established anchor:** a 2-layer 427k-parameter transformer groks the full variable-base DLP mapping at p=113 after a prolonged memorization phase.
- **Scaling program:** current results span 90k to 14.4M parameters, with active work to turn the existing capacity/data observations into a controlled phase diagram with matched data and multi-seed replication.
- **Structural controls:** prime-order DLP also groks, and a disjoint-generator split reaches full held-out generalization, showing that the phenomenon is not confined to one smooth composite-order setup.
- **Problem-size scaling:** raw DLP has grokked beyond p=113, including p=521; the next methodological priority is to disentangle genuine mathematical scaling from tokenizer/model-size growth.

Possible Teddy contribution - symbolic regression of the learned algorithm

Fit sparse symbolic expressions to checkpoint activations using analysis-only group coordinates. Candidate targets include cyclic coordinates, inverse relations and output-logit structure. Compare pre-grok, transition and post-grok checkpoints, then test whether the discovered expressions transfer across model sizes. The goal is to move from "this variable is decodable" to "this compact formula approximates the computation being carried out."

Paper 2 - Grokking as Mathematical Representation Discovery

Representation-Relative Grokking and the Representational Sufficiency Framework

Working hypothesis

Delayed generalization is partly a representation-discovery cost. A task may be information-theoretically solvable from the raw input, yet hard for gradient descent until the network discovers a coordinate system in which the solver becomes simple. Explicitly supplying a task-sufficient aligned representation should therefore change the learning regime.

Representation regime	Information	Expected learning behavior
Insufficient	$H(Y R) > 0$	Full target cannot be recovered, regardless of training time.
Sufficient but difficult	$H(Y R) = 0$	Possible long memorization / delayed grokking because useful coordinates must be discovered.
Sufficient and aligned	$H(Y R) = 0$	Rapid structured generalization becomes possible because discovery cost is reduced.

Current experimental evidence

- **CRT-7 and CRT-16 are deliberately incomplete representations:** they preserve only one component of the target and stay below their information-theoretic full-target ceilings.
- **CRT-BOTH is task-sufficient:** across the current confirmatory runs it reaches essentially perfect accuracy and learns far faster than the raw representation.
- **SCRAMBLED control:** randomly relabeling the component values preserves the acceleration across five seeds, suggesting that compact factorized sufficiency may matter more than human-readable algebraic labels.
- **Broader interpretation:** the project is evolving from a CRT observation into a general theory of how representation sufficiency, alignment, discovery cost and solver complexity jointly determine neural learnability.

Where symbolic regression becomes especially interesting

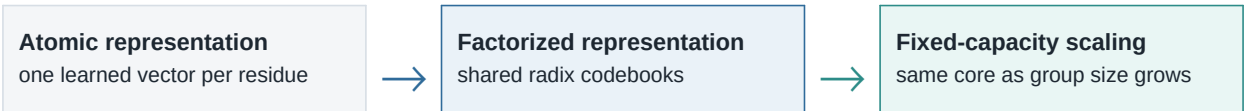
Stage	Question for symbolic regression
Before grokking	Are there weak, partial formulas already present inside the residual stream, or only example-specific structure?
Transition	Which compact equations appear or simplify immediately before generalization? Can changepoints in formula complexity predict the accuracy jump?
After grokking	Can a small symbolic program reproduce the relevant sub-computation, and does causal ablation of its supporting subspace destroy the corresponding mathematical capability?
Across models	Do different networks converge to algebraically equivalent formulas even when their raw activations or spectral routes look different?

Paper 3 - ECDLP Grokking and Fixed-Capacity Scaling

From successful toy-curve grokking to a curve-size-independent neural representation

Current status

- **Multiple ECDLP / EI4 curve configurations have already grokked.** The task is variable-base relative ECDLP: given $A=[a]G$ and $B=[b]G$, predict x such that $B=[x]A$.
- **The key scaling confound is now understood:** an atomic residue vocabulary gives every field element its own trainable vector, so embedding and output overhead grow with the curve / field size.
- **Current solution:** Factorized Residue Embedding builds each residue compositionally from a fixed small radix codebook instead of allocating one free vector per residue.
- **First FRE results are positive:** the same ~425k core has grokked $r=251$ and $r=509$, showing that one-vector-per-residue lookup memory is not necessary for EI4 grokking.
- **Next objective:** keep the trainable parameter budget independent of curve size and map the fixed-capacity frontier as r grows.



Proposed collaboration role across the three papers

Workstream	Concrete contribution
Symbolic-regression MI	Search for compact formulas mapping hidden activations to analysis-only mathematical coordinates; track formula complexity through memorization, grokking and cleanup.
Cross-scale validation	Test whether the same symbolic relations survive changes in model size, data regime, group order and surface representation.
Causal verification	Use discovered formulas to define targeted subspaces/interventions, then distinguish descriptive fits from computations the network actually relies on.
Theory & framing	Help connect empirical grokking behavior to symbolic regression, structure-aware ML, geometric/topological perspectives and broader questions about representation-dependent learnability.

How I would like to start

I would be very happy to explore fluid dynamics together and already have ideas for projects there. In parallel, these three grokking papers offer a concrete way to begin collaborating immediately. The DLP paper is the most mature; the representation-discovery and ECDLP papers have more room for a collaborator to shape the methods, analysis and theoretical framing from the start. Co-authorship would follow substantive research contribution rather than being a nominal addition.

Suggested first step: choose one mechanistic question from the DLP or representation-discovery work, define a symbolic-regression pipeline together, and test whether it recovers a compact mathematical description that is predictive, stable across seeds, and causally relevant.